

Veloura: AI Powered Virtual Try-On System

1 Problem Understanding

Clear Understanding of the Problem

Goal

Generate photorealistic images of a target person wearing a chosen garment (product image → person image) while preserving:

- Person's identity
- Pose and body structure
- Fine garment textures

This allows e-commerce users to **virtually try on clothes**, reducing uncertainty during online purchases and potentially lowering return rates.

1.1 Core Problem Elements

Input

- RGB image of a person
- Image of a target garment (product photo)

Output

A realistic synthesized image of the same person wearing the selected garment while preserving:

- Pose and body shape
- Garment geometry (sleeves, collars, hems)
- Fine textures and patterns (logos, prints, folds)
- Natural occlusions (hands over clothing, layered garments)

Constraints

- Single-view inputs (no 3D scans)
- Varied human poses
- Occlusions and overlapping clothing
- Real-world variations in lighting and backgrounds

1.2 Why This Problem is Challenging

- **Deformation and Alignment** Clothing must be warped to match the person's pose and body shape.
- **Texture Preservation** Fine patterns and fabric details must remain realistic after geometric transformations.
- **Occlusion Handling** The system must handle situations where body parts or other garments overlap the target clothing.
- **Resolution vs Speed Tradeoff** High-quality generative models such as diffusion models provide realism but require high GPU resources.
- **Generalization** Models trained on curated datasets often fail when applied to real-world images.

2 Proposed Solution

High Level Solution

Build a **hybrid virtual try-on system** combining:

- Precise geometric alignment using **flow-based warping**
- High visual realism using **latent diffusion models**
- Efficient inference using **lightweight refinement and model distillation**

2.1 Core Pipeline

1. Use a **flow-based warping module** to align garment geometry to the person.
2. Use a **conditional latent diffusion model** to generate photorealistic pixels.
3. Apply a **lightweight refinement network** to correct local artifacts.
4. Use **knowledge distillation** to compress the model for faster inference.

3 Technical Approach

3.1 1. Preprocessing and Person Representation

- Human parsing to generate semantic maps of body regions.
- Pose estimation using keypoints (e.g., OpenPose).
- Generate clothing-agnostic person representation by masking torso region.

These preprocessing steps follow methods used in **VITON** and **VITON-HD** systems.

3.2 2. Garment Warping

- Use an appearance-flow estimator to map garment pixels to body coordinates.
- Inspired by methods such as **ClothFlow** and **GP-VTON**.
- Output includes:
 - Warped garment image
 - Occlusion or confidence map

3.3 3. Latent Conditional Synthesis

- Encode the warped garment and person representation into latent space.
- Use a **latent diffusion model** conditioned on:
 - Garment latent representation
 - Person latent representation
 - Image prompt adapter for texture preservation
- Inspired by **LaDI-VTON** and **IDM-VTON** approaches.

3.4 4. Refinement and Compositing

- Use a lightweight **U-Net** or **GAN refinement network**.
- Fix seams, texture inconsistencies, and visual artifacts.
- Blend synthesized clothing region into the original background image.

3.5 5. Distillation and Efficiency

- Train a smaller student network to mimic the teacher diffusion model.
- Apply knowledge distillation for faster inference.
- Optional quantization for additional efficiency.

3.6 6. Optional 3D or UV Mapping

If computational resources allow, integrate **Pix2Surf** inspired **UV mapping** to improve texture mapping for structured garments.

4 Datasets and Evaluation

4.1 Datasets

- **VITON / VITON-HD** datasets for paired garment-to-person training.
- **DressCode dataset** for additional garment diversity.
- Other curated fashion datasets used in virtual try-on research.

4.2 Evaluation Metrics

- **LPIPS** – perceptual similarity
- **FID** – realism of generated images
- **SSIM** – structural similarity
- **Alignment Score** – IoU or keypoint alignment metrics

Human evaluation will include:

- A/B comparison between real and synthesized images
- User preference and purchase-intent surveys

5 Implementation Plan

Week 1 – Setup and Baseline

- Run VITON or ClothFlow baseline
- Prepare dataset pipeline
- Implement parsing and keypoint extraction

Week 2 – Warping Module

- Implement appearance-flow warping network
- Visualize garment alignment

Week 3 – Latent Diffusion Prototype

- Integrate latent diffusion framework
- Train conditioned on garment and person latents

Week 4 – Refinement and Distillation

- Add refinement network
- Begin model distillation for speed

Week 5 – Evaluation

- Compute LPIPS, FID, SSIM
- Conduct small user study

Week 6 – Finalization

- Final model checkpoints
- Prepare GitHub repository
- Build simple Flask demo for virtual⁹ try-on